

Hypothesis 1 - Effect of Emotional Condition on Working-Memory Performance

Confirmatory Hypothesis Report

Hypothesis

Positive, negative, and neutral emotional conditions differentially affect working-memory performance.

Method

A trial-level confirmatory analysis was conducted to evaluate whether working-memory performance differed across neutral, positive, and negative emotional conditions. The analytical sample included 23 participants.

Response accuracy was analyzed using a generalized estimating equation (GEE) model with a binomial distribution, logit link, exchangeable within-participant correlation structure, and robust standard errors. Emotional condition was entered as the primary predictor, with the neutral condition specified as the reference category. The accuracy model included 17,820 trial-level observations.

Reaction time was analyzed among valid correct-response trials only. Because reaction-time data were positively skewed, reaction time was log-transformed prior to modeling. A linear mixed-effects model was fitted with emotional condition as a fixed effect and a participant-specific random intercept. The reaction-time analysis included 14,184 observations. Maximum-likelihood estimation was used.

Model coefficients from the binomial GEE were exponentiated and interpreted as odds ratios. Coefficients from the log reaction-time model were exponentiated and converted to percentage changes relative to the neutral condition. Statistical significance was evaluated using a two-sided alpha level of .05, and 95% confidence intervals are reported.

Results

Accuracy

Descriptively, accuracy was highly similar across emotional conditions. Mean trial-level accuracy was 83.8% in the negative condition, 85.0% in the neutral condition, and 85.1% in the positive condition.

Relative to the neutral condition, the negative condition was associated with a small, non-significant reduction in the odds of a correct response (beta = -0.084, SE = 0.064, $z = -1.31$, $p = .189$), corresponding to an odds ratio of 0.92 (95% CI [0.81, 1.04]).

The positive condition did not differ significantly from the neutral condition (beta = 0.002, SE = 0.057, $z = 0.03$, $p = .979$), with an odds ratio of 1.00 (95% CI [0.90, 1.12]).

Taken together, these findings provide no evidence that emotional condition substantially altered the probability of producing a correct response.

Table 1. Descriptive accuracy by emotional condition.

Condition	Participants	Trials	correct	Accuracy	SD	Median
Negative	22	5940	4978	83.8%	0.368	1.0
Neutral	23	5940	5049	85.0%	0.357	1.0
Positive	22	5940	5055	85.1%	0.356	1.0

Table 2. GEE estimates for response accuracy.

Contrast	OR	95% CI	p
Negative vs Neutral	0.92	[0.81, 1.04]	= .189
Positive vs Neutral	1.00	[0.90, 1.12]	= .979

Reaction Time

In contrast to accuracy, emotional condition was strongly associated with reaction time.

Relative to the neutral condition, the negative emotional condition was associated with a significantly lower log reaction time (beta = -0.265, SE = 0.015, $z = -17.29$, $p < .001$). After back-transformation, this coefficient corresponded to an estimated 23.3% reduction in reaction time relative to the neutral condition.

The positive emotional condition was also associated with a significantly lower log reaction time compared with the neutral condition (beta = -0.285, SE = 0.015, $z = -18.60$, $p < .001$). This corresponds to an estimated 24.8% reduction in reaction time relative to neutral.

The magnitude of the positive and negative coefficients was similar, suggesting that both emotionally induced conditions were associated with faster responses compared with the neutral baseline.

Table 3. Descriptive reaction time by emotional condition.

Condition	Participants	Trials	Mean RT	Median RT	SD
Negative	22	4704	740.25	433.50	744.23
Neutral	23	4829	966.58	630.00	896.36
Positive	22	4651	771.33	421.00	808.86

Table 4. Mixed-effects model estimates for log reaction time.

Contrast	Beta	RT change	95% CI	p
Negative vs Neutral	-0.265	-23.3%	[-25.5%, -20.9%]	< .001
Positive vs Neutral	-0.285	-24.8%	[-27.1%, -22.5%]	< .001

Decision Rules

- H1 is supported if there is clear evidence of differences between emotional conditions in accuracy, reaction time, or both.
- H1 is partially supported if differences arise in only one outcome or in only one specific contrast.
- H1 is not supported if there is no evidence of relevant differences between conditions.

Interpretation

Hypothesis 1 was partially supported.

The emotional manipulation did not produce statistically detectable differences in response accuracy. Accuracy remained approximately 84-85% across the three emotional conditions.

However, emotional condition showed a pronounced association with reaction speed. Both positive and negative emotional conditions were associated with substantially faster responses than the neutral condition.

Importantly, the faster responses under emotional induction were not accompanied by a statistically significant loss of accuracy. Thus, the results suggest that emotional induction influenced the speed component of working-memory performance more strongly than the probability of a correct response.

Because both positive and negative conditions showed effects in the same direction relative to neutral, these results do not demonstrate a valence-specific advantage of positive emotion or a valence-specific impairment under negative emotion. Those directional predictions are evaluated in Hypotheses 2 and 3.

Hypothesis Assessment

Conclusion: Hypothesis 1 was partially supported. Emotional condition did not significantly affect accuracy, but both positive and negative emotional conditions were associated with substantially faster reaction times relative to the neutral condition.